



Cardiac adaptation to exercise training in health and disease

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Abstract

The heart is the primary pump that circulates blood through the entire cardiovascular system, serving many important functions in the body. Exercise training provides favorable anatomical and physiological changes that reduce the risk of heart disease and failure. Compared with pathological cardiac hypertrophy, exercise-induced physiological cardiac hypertrophy leads to an improvement in heart function. Exercise-induced cardiac remodeling is associated with gene regulatory mechanisms and cellular signaling pathways underlying cellular, molecular, and metabolic adaptations. Exercise training also promotes mitochondrial biogenesis and oxidative capacity leading to a decrease in cardiovascular disease. In this review, we summarized the exercise-induced adaptation in cardiac structure and function to understand cellular and molecular signaling pathways and mechanisms in preclinical and clinical trials.

Keywords Exercise · Heart · Cardiac hypertrophy · Cardiac adaptation · Cardiovascular disease

Introduction

The heart is a vital organ in our body that works continuously to effectively pump blood throughout the body and to supply the peripheral organs with essential elements, such as oxygen and

nutrients [85]. This workload accounts for approximately 20% of the total oxygen consumption in the body and is performed by the cardiac contraction system to maintain maximum pump efficiency and cellular metabolism [70]. Furthermore, maximal exercise induces a tenfold increase in oxygen consumption by the heart [85], indicating a metabolic adaptability to exercise [93]. An “athletic heart” is a form of adaptation to chronic exercise training in which cardiac remodeling occurs in response to an increased workload by undergoing cardiac hypertrophy, expanding in size without the proliferation of cardiomyocytes. This physiological cardiac hypertrophy is usually accompanied by increased mitochondrial energy production [26]. Physiological cardiac hypertrophy is associated with an improvement in normal cardiac contractile function at rest, whereas pathological cardiac hypertrophy is associated with a deterioration of cardiac contraction and energy metabolism [69]. Cardiac hypertrophy has been quantified by measuring changes in cardiac wall thickness [44]. However, cardiac wall thickening may occur without any changes to the ventricular volume. Based on the patterns of cardiac wall ventricular thickening and volume, cardiac hypertrophy is classically characterized into cardiac concentric or eccentric hypertrophy [14, 105]. Both physiological and pathological cardiac hypertrophy reflect changes in wall thickening and ventricular volume, while pathological cardiac hypertrophy is characterized by severe cardiac remodeling that ultimately leads to heart failure (HF) and dysfunction (Fig. 1) [79].

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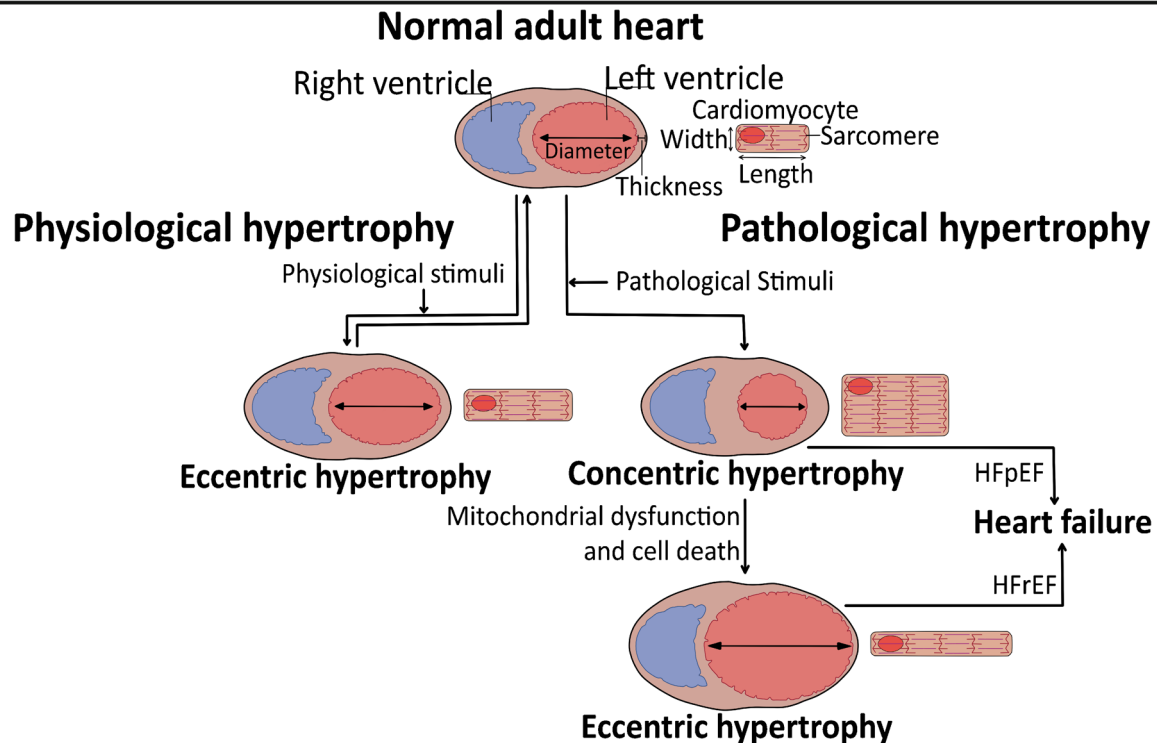


Fig. 1 Summary of physiological and pathological cardiac hypertrophy. Solid arrows indicate proven pathways. Eccentric hypertrophy (physiological): preserved, mildly enhanced cardiac function; Concentric hypertrophy (pathological): compensated, preserved contractile function; Eccentric hypertrophy (pathological): maladaptive

remodeling, contractile function; HFpEF: heart failure with preserved ejection fraction; HFrEF: heart failure with reduced ejection fraction. Reproduced Nakamura et al. [79] with permission of Springer Nature Copyright Clearance Center's RightsLink

Regular exercise training is known to be an effective intervention that improves the quality of life (QoL) and extends life expectancy [6, 78]. Exercise training, particularly aerobic exercise, plays a key role in reducing morbidity and mortality and improving the conditions of patients with cardiovascular diseases [49]. Although the advantages of regular exercise training are multifactorial and reflect the integrated adaptations in various organ systems [50, 93], many researchers have not determined the extent to which exercise account for specific cellular and molecular mechanisms responsible for reversing cardiac contractile dysfunction and various metabolic disorders.

A clear distinction in the upstream signaling pathways and the resulting transcriptional responses have been observed between physiological and pathological cardiac hypertrophy in the cellular and molecular levels [83]. This review focuses on the cellular, molecular, metabolic, and morphological cardiac adaptations to chronic exercise training. First, we will review a “systematic” approach to defining cardiac adaptation to exercise training in the cardiovascular system. We will discuss whether exercise training contributes to the development of physiological and pathological cardiac hypertrophy. Additionally, we will discuss the

morphological characteristics of cardiac hypertrophy and its effect on the progression of heart disease and failure. Second, we will provide a general explanation of the cellular and molecular events that cause physiological cardiac adaptation. Studies have described various molecular mechanisms that mediate the cardiac adaptation to exercise training. For example, insulin-like growth factor 1 (IGF-1) and other downstream signaling pathways are key mediators of physiological cardiac hypertrophy. Moreover, accumulating evidences from mouse genetic models have proven that physiological cardiac hypertrophy is directly antagonistic to pathological cardiac hypertrophy. Finally, the cellular and molecular mechanisms involved in the cardiac adaptation to exercise training will be examined, followed by a review of methods for identifying new therapeutic approaches to treat HF.

Cardiac adaptation to exercise training

The cardiovascular system reflects noticeable adaptations to acute and chronic exercises. The core of these adaptations involves the cooperative integration of changes in the cellular and molecular signaling pathways within the cardiovascular

system that allow the tissue to respond to the physical demands generated by exercise training [122]. For example, when cardiac output (CO) per unit time is measured during acute exercise, the cardiac contractility increases by four to eightfold [122]. Changes in cellular signaling associated with the activation of sympathetic and parasympathetic nerves are the basis for the increase in cardiac contractility, which has an immediate influence on the heart rate (HR) and CO [76]. As a closed network, the cardiovascular system increases blood flow, consequently causing an increase in blood pressure (BP) and redistribution of oxygen and nutrients to the exercising skeletal muscles. Despite the initial vascular resistance, physical exercise induces vasodilation, allowing for the increased inflow of oxygen-rich blood to be distributed to the exercising skeletal muscles. This local signaling activates a blood redistribution mechanism in which the cellular signaling pathways within the vascular system facilitate a tenfold increase in the blood flow to the exercising skeletal muscles [39].

Acute and chronic exercise differentially influences exercise-induced cardiovascular adaptations. Acute exercise-induced cardiac response reflects the remarkable reserve capacity of the cardiovascular system. During acute exercise, the hematological adaption resulting from hemodynamic changes in the peripheral blood flow leads to increased instant oxygen supply and metabolic adaptation [47]. In contrast, long-term exercise training, which involves endurance/dynamic training or power/strength training, brings about chronic changes in the circulatory system by inducing physiological and morphological adaptations [92]. For example, dynamic sports, such as running, demand a continuous increase in cardiac function, while static sports, such as powerlifting, require an instantaneously high BP. Meanwhile, cycling and rowing exercises continuously induce both dynamic and static conditions during exercise. Physiological demands are accepted by the cardiovascular system as it also adapts to the more dominant form of hematological stress, depending on the type of exercise (e.g., aerobic, resistance). The physiological or morphological changes induced in the circulatory system may determine the categorization of the types of exercise [34]. Generally, long-term exercise training strengthens aerobic ability of the skeletal muscles and reduces CO at rest. These changes are attributed to a slow HR (bradycardia) and increased myocardial contractility. In contrast, muscle strength training causes a rapid increase in BP and HR, but it may not cause any changes in basal cardiac function [68, 92]. Moreover, muscle strength training causes skeletal muscle growth; therefore, an increase in the skeletal muscle cross-sectional area leads to a higher overall oxygen perfusion to the skeletal muscles. Thus, the circulatory system may possess the innate ability to differentiate exercise types according to the differences in hematological stress. Therefore, cardiac hypertrophy induced by exercise does not pertain to the number

of cardiomyocytes but is generally accepted as an increase in the size of the heart.

Adaptive cardiac remodeling, a characteristic of physiological cardiac hypertrophy, is classified as concentric or eccentric (Fig. 1) [41]. To verify these types of cardiac remodeling, many researchers consider on the geometric relationship between the ventricular size and ventricular wall thickness, or relative wall thickness (RWT) [32]. Concentric cardiac remodeling is characterized by an increase in heart size due to uneven increases in ventricular volume and wall thickness, often manifested in patients with heart disease. In contrast, eccentric cardiac remodeling refers to an increase in heart size, while RWT refers to an organically adjusted increase in both ventricular volume and wall thickness [129]. Hematological stress induced by dynamic (running) or static (muscle strength) exercise reflects HR, which is an important determinant of cardiac function. For example, dynamic exercise training increases cardiac contractility through the Starling effect, in which the increased amount of venous blood returning to the heart ultimately increases the amount of blood released from the heart. This resulting chronic “volume overload” induces eccentric cardiac remodeling [126]. In contrast, muscle strength exercise induces increases in BP by exerting sudden pressure, which causes heartbeat impedance to increase, resulting in concentric cardiac remodeling.

Dynamic (intermittent) changes, in both the structural and geometric adaptations of the heart to exercise training, are differentiated from chronic hematological stress, such as hypertension [61]. Regardless of the type of hypertrophic adaptation, exercise-induced cardiac remodeling is considered a typical form of adaptation as HF does not occur. Exercise-induced physiological cardiac hypertrophy leads to improvement in cardiac function and adaptive geometry and, thus, is a process fundamentally different from hypertension-induced pathological cardiac hypertrophy [1, 62]. In addition, cardiac hypertrophy induced by exercise training is completely reversible, without any negative influences [69, 110]. In contrast, cardiac hypertrophy caused by cardiac disease, gradually induces a loss of normal cardiac function as the tissues shift from concentric growth with an elevated RWT to a dilated state with a decreased RWT [35, 42]. However, because fundamental cellular changes, such as fibrosis, are unable to be resolved, researches have not clearly determined whether the reversal of these changes is sustainable [36].

Gene regulatory mechanisms underlying the molecular and metabolic adaptations in exercise-induced cardiac hypertrophy

Accumulation of studies has reported on cardiac transcriptomic changes after exercise training [17, 23, 114]. Based on these results, exercise training or pathological

stimulation leads to different adaptation of gene expression in cardiac structure and function. These results also provide a comparison of the transcriptomic differences between pathological and physiological cardiac hypertrophy [115]. Activation of fetal gene expression, which includes the induction of natriuretic peptides and fetal sarcomeric isoform gene expression, is a characteristic of pathological cardiac hypertrophy. This pattern, however, is not usually observed in the motor transcriptome [115]. For example, swimming exercise and aortic banding (pathological hypertrophic stimulation) produced noticeable changes in gene expression patterns [72]. In contrast to the distinct changes in the expression of arterial and brain natriuretic peptides (ANP and BNP) observed following banding, these changes did not appear following swimming exercise. Similarly, the expression of fetal β -myosin heavy chain (MHC) was clearly upregulated by aortic stenosis, but not by swimming [72]. Metabolic gene expression patterns are also clearly differentiated in pathological and physiological cardiac hypertrophy. Gene expression patterns related to the pathways of energy metabolism and bioenergy are altered in response to pathological myocardial hypertrophy and reprogramming established during cebelitas cordis. This metabolic reprogramming of the mechanism regulating gene expression is reminiscent of the “fetal” change that reduces the ability to undergo fatty acid oxidation and downregulates the expression of nuclear and mitochondrial genes involved in mitochondrial energy delivery and respiratory pathways [7]. In contrast, some genes involved in the glucose uptake pathways are upregulated. This energy metabolism reprogramming may contribute to the onset of HF. Changes in the levels of energy metabolism genes are particularly interesting and important. In healthy adult hearts, the expression of genes involved in mitochondrial fatty acid oxidation (FAO), the primary source of adenosine triphosphate (ATP), is downregulated in the diseased state of pathological cardiac hypertrophy [7], whereas gene expression does not change in response to exercise [9]. Moreover, the expression of cardiac genes that encode members of the pathways associated with lipid metabolism, fatty acid transport, and fatty acid-binding proteins is increased because of exercise, but not in the presence of pathological cardiac hypertrophy or HF [98, 115]. Such regulatory gene reprogramming is also reflected in the fuel oxidation rate. In particular, palmitate oxidation rates, a measurement of FAO, are increased in the hearts of trained rats [18], whereas the FAO is decreased in pathological cardiac hypertrophy and HF models [121]. Following exercise training, the heart exhibits a mitochondrial biogenic adaptation, but the oxidative ability is reduced in diseased hearts. The levels of the factors involved in the electron transport chain, mitochondrial DNA (mtDNA), and citrate synthase activator elements are all increased in the hearts of the mice that underwent swimming exercises [98]. Mitochondrial respiration is also increased in the hearts of

the trained mice when palmitoylcarnitine is used as the substrate, consistent with changes in the expression of fatty acid-consuming genes [84]. Quantitative measurements of the metabolites in major energy metabolism pathways reveal correlations between the changes in the expression of cardiac metabolic genes and responses to exercise and pathological stimulations [65]. Measurement of the intermediate products of metabolites using metabolomics may represent the characteristics of the changes in energy metabolism, similar to the way transcriptomics represents changes in gene expression following exercise training [45]. Some studies reported the measurements of metabolites in patients and animals with HF, showing changes in acylcarnitine levels that reflect changes in mitochondrial FAO [8, 64, 100]. Interestingly, comparisons between HF and preclinical models of exercise revealed noticeable differences. Acylcarnitine levels increase in the initial stages of pressure-overload cardiac growth/myocardial infarction models; however, mice that perform voluntary wheel running exercises exhibit low acylcarnitine levels [64]. The intermediates of the tricarboxylic acid (TCA) cycle also exhibit a similar pattern. In summary, these results reveal distinct differences in the metabolic reprogramming between pathological cardiac hypertrophy and exercise-induced physiological cardiac hypertrophy, in conjunction with the changes in gene expression. The maintenance of FAO and ATP production is consistent with the adaptive responses to increased workloads. In contrast, pathological cardiac hypertrophy causes low levels of fetal programming, which in turn causes inappropriate energy responses.

More recent studies have shown that peroxisome proliferator-activated receptor gamma coactivator 1- α (PGC-1 α), which leads to mitochondrial biogenesis for beige adipocytes, is a key regulator of exercise-induced mitochondrial adaptation in the heart and skeletal muscle [29, 94]. For example, acute exercise induces an activation of PGC-1 α expression in skeletal muscle [91], but PGC-1 α and PGC-1 β knockout mice exhibit reduced mitochondrial function and exercise performance [130, 133]. Consistent with the animal study results, exercise training also promotes mitochondrial synthesis, palmitate oxidation, and changes in mtDNA in the heart [84, 98]. However, additional studies using a heart-specific PGC-1 α knockout animal model are still needed to verify the roles of PGC-1 α in mitochondrial and energy metabolic adaptations to exercise observed in mice. The activation of PGC-1 α and interactions with downstream cascades of transcription factors, including nuclear receptors, mediate the numerous actions of PGC-1 α in the heart [121]. In addition, some studies suggest that PGC-1 α binding partner and coactivator PPAR- α regulates fatty acid oxidation in the heart [31, 38, 46]. In pathological cardiac growth and HF conditions, downregulated PGC-1 α and PPAR- α signaling alters the energy metabolism [7]. Moreover, PGC-1 α facilitates the mitochondrial synthesis response by interacting with both the

estrogen-related receptor (ERR) and nuclear receptor factor 1 (NRF-1) [103]. ERR α , a major isoform of ERR, is required for the adaptation to BP overload and peak ATP synthesis [51]. However, the exact role of ERR in the cardiac adaptation to exercise training remains to be identified. In addition to PGC-1 α , unbiased profiling studies have identified a knock-down of C/EBP β , namely, cardiac binding protein β . Chronic swimming exercise in the heart leads to downregulated C/EBP β expression. Intriguingly, C/EBP β knockdown causes changes in cardiomyocyte growth and gene activation that are analogous to the changes induced by exercise [15]. Concurrently, these results increase levels of CBP/p300-interacting transactivator CITED4, which contributes to exercise-induced cell proliferation in the heart [15]. Based on these observations, a recent work has suggested that CITED4 overexpression is positively associated with the activation of fetal genes and protection of ischemia reperfusion injury [11]. However, the role of exercise-induced CITED4 mechanism remains poorly understood.

Noncoding RNAs are associated with the molecular regulatory mechanisms that also contribute to cardiovascular health and disease [118]. Exercise training also regulates the expression of various miRNAs in the heart [37, 67]. To date, these miRNAs have been shown to regulate the genes involved in various aspects of cardiac structure and function. For example, miR-29 suppresses the expression of fibrotic genes and exercise training has been found to activate miR-29 expression levels in cardiac fibroblasts [112, 119]. Recent evidence suggests that miR-222 is a key contributor for physiological cardiac growth following exercise training, resulting in the physiological hypertrophic phenotype of cardiomyocytes in vitro by increasing the α -MHC/ β -MHC ratio and by suppression of ANP and BNP. Moreover, miR-222 is essential for physiological growth in vivo and protects the heart against ischemic injury [67]. The studies are summarized in Table 1.

Cellular signaling pathways involved in the cardiac hypertrophy to exercise

Exercise training regulates molecular adaption and cellular signaling pathways through downstream transcriptional factors and other regulatory mechanisms in the heart. To identify exercise-induced signaling pathways for physiological cardiac hypertrophy, many researchers have used genetic mouse models. McMullen et al. [72] reported that the PI3K/Akt pathway is a major mediator for physiological growth and PI3K is essential for exercise-induced hypertrophy. This is in agreement with the previous study, which revealed that exercise training protects against pathological cardiac hypertrophy through the activation of PI3K [128]. Much research has focused on whether exercise is able to activate the PI3K signaling pathway. Brain et al. [28] suggest that Akt, known as the

downstream kinase of the PI3K signaling pathway, plays a crucial role in the enhancement of physiological cardiac hypertrophy. In addition, Akt1 promotes physiological cardiac hypertrophy through altered aortic constriction, showing antagonistic pathological hypertrophy response.

The PI3K/Akt pathway released from IGF-1 and insulin is a main regulator for cardiac hypertrophy. Seneri et al. [108] have investigated and found that exercise training increases the levels of cardiac IGF-1 in athletes. The IGF-1 receptor (IGF-1R) is indispensable in inducing physiological hypertrophy in animals [57, 73]. Cardiac-specific, insulin receptor-deficient mice show decreased ATP production and mitochondrial respiration [16]. Surprisingly, these results did not cause a change in PGC-1 α levels, although the PPAR- α expression was decreased [16]. However, exercise training induces the activation of PGC-1 α and the mitochondrial capacity in insulin receptor-deficient mice [98]. These results verify the important roles of IGF-1 and insulin-stimulation signaling pathways for cardiac growth. Several studies have revealed that other cellular signaling mechanisms influence and activate the PI3K/Akt pathway. For example, the growth factor neuregulin-1 induces ErbB2/ErbB4 tyrosine kinase receptor activation, resulting in the activation of PI3K signaling [25, 43]. In particular, the neuregulin-1/ErbB2/ErbB4 pathway also regulates cardiomyocyte proliferation and differentiation [10, 25]. In line with similar studies, exercise training induces the upregulation of cardiac neuregulin-1 expression [21, 127] and expression of cardiomyocyte markers [15, 21]. Nevertheless, the precise role of neuregulin-1 signaling associated with exercise, including cardiomyocyte proliferation, is yet to be determined.

Nitric oxide (NO) mediates the advantageous effects of exercise on the heart. This result indicates that NO generated by endothelial nitric oxide synthase (eNOS) activates soluble guanylate cyclase and increases the activities of both cyclic guanosine monophosphate and protein kinase G [132]. Another possibility is that exercise training also increases circulating NO levels mediated by increased eNOS, resulting in protection from ischemia/reperfusion injury [22]. However, these benefits of exercise after myocardial infarction were observed in the eNOS knockout animal models [27]. Results indicated that exercise training induces beta 3-adrenergic receptor and prevents against chronic adrenergic activation through the activation of eNOS [132]. Finally, swimming exercise also increase mitochondrial synthesis and eNOS expression, resulting in cardiac remodeling in eNOS-deficient mice [123]. The studies are summarized in Table 2.

Animal models of physiological cardiac hypertrophy

There is growing evidence that exercise training could be associated with an increased risk of physiological cardiac

Table 1 Summary of well-studied genes involved in exercise-induced cardiac hypertrophy.

Subjects	Exercise type	Genes	Regulation	Function	References
Rat	Treadmill (60 min, 6 days/week, for 8 weeks)	ANP	↑	Improves endocrine structure and function	[87]
Human	4 consecutive days of walking exercise (30–50 km/day)	BNP	↑	Increases myocardial stretch	[2]
Rat	Voluntary wheel running for 6 weeks	α -MHC β -MHC	↑ ↓	Protects cardiotoxicity	[52]
Mouse	Treadmill (60 min/day, days/week for 15 weeks)	PGC-1 α	↑	Increases in mitochondrial biogenesis	[124]
Rat	Swimming (60 min/day, 5 days/week for 8 weeks)	PPAR- α	↑	Enhances energy metabolism regulation	[102]
Mouse	Swimming (90 min, two times daily for 14 days)	C/EBP β	↓	Improves cardiomyocyte growth and proliferation	[15]
Mouse	Treadmill (60 min, 5 days/week for 4 weeks)	CITED4 Myonectin	↑ ↑	Ameliorates acute myocardial ischemic injury by suppressing apoptosis and inflammation	[86]
Rat	Swimming (60 min, days/week for 10 weeks)	miR-29c	↑	Ventricular compliance	[112]
Mouse	Swimming (90 min, days/week for 10 weeks)	miR-17-3p	↑	Inhibits myocardial ischemia-reperfusion injury	[109]
Mouse	Swimming (90 min, twice daily, 6 days/week for 4 weeks)	miR-222	↓	Improves cardiomyocyte growth and proliferation	[67]
Rat	Swimming (60 min, 5 days/week for 10 weeks)	miR-126	↑	Increases cardiac angiogenesis	[111]
Rat	Swimming (60 min, 5 days/week for 10 weeks)	miR-208a miR-208b	↓ ↑	Improves epigenetic regulation	[113]

ANP, arterial natriuretic peptides; BNP, brain natriuretic peptides; α -MHC, alpha myosin heavy chain; β -MHC, beta myosin heavy chain; PGC-1 α , peroxisome proliferator-activated receptor gamma coactivator 1-alpha; PPAR- α , peroxisome proliferator-activated receptor alpha; C/EBP β , a member of the CCAAT/enhancer-binding protein beta; CITED4, Creb-binding protein [CBP]/p300-interacting transactivator with ED-rich carboxy-terminal domain-4; miR, micro RNA

hypertrophy [71]. This problem is particularly poignant in subjects with hypertrophic cardiomyopathy (HCM), a genetic disease with a prevalence of 0.2–0.5%. The exercise-induced cardiac hypertrophic phenotype overlaps with that of HCM, or the differences may be negligible, and thus are considered indistinguishable by cardiac geometry measurements. In most cases, HCM progresses to dilated cardiac dysfunction, but may lead to sudden cardiac death. HCM accounts for 35–50% of the cases of sudden cardiac death among young athletes [24]. Therefore, more studies are required to understand the differences in the molecular mechanisms that lead to HCM between subjects undergoing exercise training and those with

the disease. Animal models have often been used to examine cardiac remodeling and adaptation in response to exercise [63]. Specific molecular and metabolic signals required for differentiating the causes of cardiac hypertrophy have been identified in these studies.

Animal models offer many practical advantages for conducting research on human conditions [106]. Genetically modified mouse models represent the foundation for identifying the cellular and molecular pathways that underlie cardiac biology. With the exception of a few differences among strains, rodents experience physiological adaptations to exercise, and the type and scope of the adaptive changes depend

Table 2 Summary of well-studied cellular signaling pathways involved in exercise-induced cardiac adaptation.

Subjects	Exercise type	Genes	Regulation	Function	References
Mouse	Swimming (90 min, 7 days/week for 4 weeks)	PI3K	↑	Induces physiological hypertrophy	[72]
Mouse	Treadmill (30 min, 5 days/week for 4 weeks)	Akt	↑	Improves myocardial ischemia-reperfusion injury	[95]
Mouse	Swimming (90 min, 5 days/week for 6 weeks)	IGF-1	↑	Stimulates NO production in mice cardiomyocytes	[19]
Rat	Treadmill (60 min, 5 days/week for 10 weeks)	IGF-1R	↑	Protects cardiac apoptosis	[66]
Mouse	Swimming (90 min, 5 days/week for 9 weeks)	NO	↑	Improves systolic function and recover pathological remodeling in failing hearts	[125]
Rat	Resistance exercise (electrical stimulation) 20 V, 0.3-s duration, at 3-s intervals, 40% 1RM	eNOS	↑	Inhibits myocardial ischemia-reperfusion injury	[101]

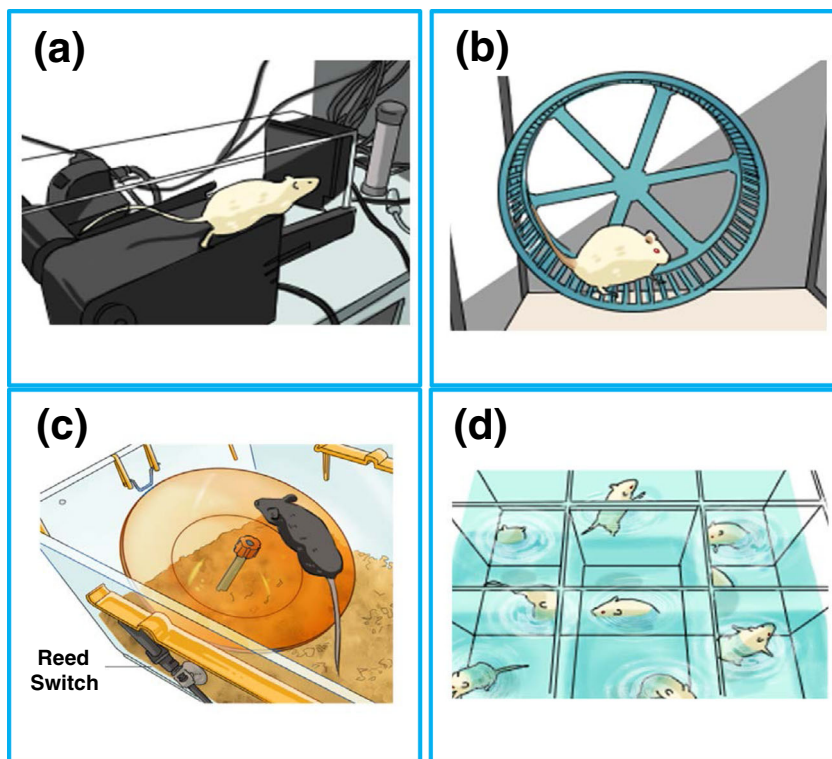
PI3K, phosphoinositide 3-kinases; Akt, protein kinase B; IGF-1, insulin-like growth factor 1; IGF-1R, he insulin-like growth factor 1 receptor; NO, nitric oxide; eNOS, endothelial nitric oxide synthase

on the type of exercise. In general, rodent exercise protocols are divided into three types: (1) running on a treadmill, (2) voluntary wheel running, and (3) swimming. Most studies use protocols based on the intensity, grade, and duration of animal treadmill running. Studies on treadmill-induced adaptations have applied endurance running protocols, in which the speed, duration, and grade were either fixed or gradually increased. Details of these protocols are comprehensively explained in many reviews (Fig. 2) [106].

In previous studies, regardless of the exercise protocol, rodents exhibited a predictable physiological adaptation to endurance exercise training that mimics human physiology [107]. Similarly to humans, rats and mice show exercise-induced cardiac hypertrophy characterized by a 5–24% increase in cardiac mass following 4 weeks of exercise training [30, 77]. For example, interval exercise training increases 30% of the cardiac growth compared with other exercise protocols in animals [48, 55]. These approaches may be suitable for patients with cardiovascular diseases, which include low-intensity exercise over a long duration (20–60 min) [89]. Many studies demonstrated a detailed mechanism for interval exercise training protocol in animals to be adapted to patients without side effects [96]. However, these prescribed exercise programs are negatively associated with improvements of cardiac function in patients with cardiovascular disease. Therefore, preclinical trials may improve our understanding of the cellular and molecular responses induced by these different exercise protocols.

Swimming is a stable form of aerobic exercise used to study the physiological cardiac adaptation to exercise. In swimming exercises, mice and rats are placed in a swimming tank for a specified amount of time [104, 116]. The major difference between swimming and treadmill exercises is based on how the exercise intensity is adjusted. When cardiac growth between swimming and treadmill exercises was compared, swimming exercises induced a 30–50% increase in cardiac mass, which was on average higher than the increase induced by treadmill exercises [4, 74]. Because the swimming and treadmill protocols involve forced exercise, the resulting cardiac growth may be difficult to categorize as completely arising from physiological growth, as stress may also play a role. Although the rats may generally comply with the exercises, situations that involve swimming or treadmill exercise block their natural activities and cause at least a certain level of stress-induced cardiac hypertrophy, which should be considered. Interestingly, the mouse heart is remarkable because it is able to recognize an intermittent, pathological overload, representative of the irregular exercise load, as a specific pathological factor [90]. Therefore, various researchers have used voluntary exercise paradigms in rodent studies [107]. Based on sufficient evidence, voluntary wheel running is not merely a simple behavior, but rather a quite complex one, as it is strongly influenced by the experimental conditions and rodent species [60, 75, 97]. Voluntary wheel running exercise represents suitable aerobic exercise for physiological cardiac hypertrophy and improves depressive behavior and neurological

Fig. 2 Overview of endurance animal exercise running protocols. **a** Treadmill running. **b** Wheel running. **c** Rotating track running. **d** Swimming. Reproduced from Seo et al. [106] with permission of Springer Nature Copyright Clearance Center's RightsLink



function in rodents [3, 58, 81]. In contrast, long-term loaded wheel running exercise did not increase cardiac hypertrophy [59, 81]. Although the beneficial effect of exercise training types is known, a variety of exercise-induced animal models play a pivotal role in determining the cellular and molecular mechanisms for cardiac adaptation in preclinical studies.

Translational and therapeutic implications

In preclinical models, exercise training induces physiological cardiac remodeling and inhibits pathological cardiac hypertrophy, suggesting an HF improvement. Particularly, many researchers provide evidences that exercise training directly affects the mechanism of physiological cardiac hypertrophy and pathological cardiac hypertrophy through alterations of gene expression levels in genetic animal models. For example, exercise training improves cardiac function in surgical and genetic mouse models of HF and hypertrophic cardiomyopathy [59, 74, 120], but other studies reported that low-intensity exercise did not exert an effect on hypertension in rats with HF [33].

Among clinical trials, the Heart Failure: A Controlled Trial Investigating Outcomes of Exercise Training (HF-ACTION) trial is the largest study of its kind, in which the safety and effectiveness of exercise were examined in 2300 patients with HF with a reduced ejection fraction (HFrEF) [82]. The study investigated the combined effects of walking, treadmill, and stationary bike exercises on percent HR maximum. According to the results of the HF-ACTION trial, the prescribed exercise training did not exhibit significant decreases in cardiovascular-related mortality rates or HF-related hospitalization rates compared with that of the general treatment group [82]. However, considerable improvements were observed in the 6-min walking test and in overall fitness, which indicated improvements in QoL [40]. These findings are significant when considering the fact that the HF-ACTION trial was conducted only in patients with HFrEF.

Some patients with HF are known to suffer from HF with a preserved ejection fraction (HFpEF). Although large-scale randomized studies on the effects of exercise in patients with HFpEF have not been conducted, some small-scale studies have been reported [117]. Retrospective analyses of these studies did not reveal an association between cardiopulmonary fitness and the QoL of patients with HFpEF, nor changes in their systolic and diastolic functions [88]. In these patients, exercise-induced improvements may not have exerted a substantial effect on cardiac function, suggesting possible peripheral effects on arterial stiffness, which can also affect the musculoskeletal system and endothelial function, and consequently, peripheral vascular resistance may have occurred. Given the relatively low number of patients who have registered for the HFpEF experiment to date, a randomized controlled trial

with a larger study population is necessary to confirm these results. Moreover, an assessment of the type of exercise that is most useful for patients with HFpEF or HFrEF is needed.

As mentioned above, all signaling pathways and factors that facilitate physiological hypertrophy exert protective effects on the heart and antagonistic actions on pathological hypertrophy in preclinical trials [11, 15, 67, 128]. According to the findings of the HF-ACTION trial, cardiac rehabilitation may be prescribed to patients with HFrEF; nevertheless, an exercise prescription presents significant challenges for these patients, as they have trouble in strictly complying with exercise prescription [82]. The subjects were generally elderly patients who had additional comorbidities that have interfered with their participation and compliance with the exercise prescription. Therefore, target therapies that facilitate exercise-induced activation of signaling pathways in the heart may be able to supplement the prescribed exercise program. However, the identification of safe and effective targets that activate the physiological cardiac hypertrophic pathways is a difficult task. The IGF-1/PI3K pathway is a proven target in preclinical models. Likewise, hormone replacement therapy is one of the well-known strategies for increasing the circulating IGF-1 level. Preclinical trials reported the benefits of growth hormones in HF patients [53, 131]. In particular, PPAR α and PPAR δ share many cardiac targets and induce glucose oxidation, indicating a well-balanced energy metabolism [20]. PPAR δ cooperates with AMP-activated protein kinase (AMPK) signaling in the skeletal muscles to mediate their responses to exercise [80]. Therefore, a selective PPAR δ ligand appears to be an attractive target for facilitating exercise mimetics effects under HF conditions. Moreover, direct AMPK activators function as exercise mimetics, as AMPK strengthens the activities of PPAR δ in the skeletal muscles. AMPK activation retains various metabolic effects that may be beneficial in patients with HF [56]. PGC-1 α , a direct target of AMPK, is directly associated with mitochondrial synthesis and energy production in the heart [54]. However, concerns regarding excess glycogen accumulation due to AMPK activation in cardiomyocytes lower than the expectations for this approach [5]. Although definitive activators of ERR have not yet been identified, it may also represent an effective target. Moreover, the mitochondrial-targeted antioxidant molecule SS-31, also known as elamipretide interacts with cardiolipin, a mitochondria-specific phospholipid [12]. Elamipretide has been shown to improve cardiac function and facilitates mitochondrial respiration in microembolization HF models [13, 99] and is currently being studied in clinical trials.

Summary and future perspectives

Over the past decade, an improved understanding of the detailed and quantifiable beneficial effects of exercise on both

cardiac structure and function has been achieved. Physical activities and the reversal of sedentary behavior are important in reducing the cardiovascular mortality rate. Most studies in this field have focused on the cardiac adaptation to exercise training. Endurance exercises increase the size of cardiomyocytes. These results occur in the absence of cardiomyocyte disarray and interstitial and vascular fibrosis, which are associated with pathological hypertrophy. Accordingly, we have observed that specific growth factors and signaling pathways regulate physiological and adaptive hypertrophic conditions, which are characterized by an increase in the cardiomyocytes size without the activation of fetal gene programs. However, our findings are still limited in that we have not yet performed any sort of trajectory analysis to ascertain the specific key gene or signaling mechanism that may be involved in determining the definitive fates of each type of cardiac hypertrophy as irreversible or reversible. Moreover, exercise training increases mitochondrial production by inducing cardiac remodeling to improve energy metabolism, which is partially regulated by PGC-1 α . The mitochondrial adaptation is regulated by the same signaling pathways involved in cardiac hypertrophy, ensuring that sufficient energy is produced to satisfy the increased demands following exercise training. Under HF conditions, these adaptations protect the pathological signaling pathways and ischemic/reperfusion injury. Although sufficient evidence shows that the activation of pathways associated with physiological hypertrophy may yield therapeutic benefits, drug efficacy, without side effects, and efficacy-related issues are unknown. Nonetheless, the potential benefits of exercise mimetics are sufficient to ensure continuing studies in this field.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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